



KAIROS

Kinetics of Aortic bioprosthesis Integrity, Risk-adapted echO Surveillance

καιρός, the right moment.

Dyania Health Hackathon 2026 · AVR Durability Challenge

Panagiotis Sotiropoulos, physician · Kostis Dovelos, PhD, engineering · Minas Papageorgiou, engineering

THE PROBLEM

The valve wears out quietly, and surveillance is one schedule for everyone

1 in 3

surgical bioprostheses deteriorated on echo over a decade, in 1,387 patients

~1/3

of scheduled five-year echoes never happen

10.8%

transcatheter valves by five years, in 2,403 patients

29

of our own 117 patients carry an externally mounted pericardial valve, withdrawn after an FDA warning

REOPERATIVE AVR, UK 1996–2019	MORTALITY
Elective	4.8%
Urgent	11.8%
Emergency	31.7%

Association across all reoperation causes, describing how urgency and mortality travel together. Not a count of preventable deaths.

Deterioration is mostly subclinical, so it is found late, when reoperation is urgent rather than planned.

THE QUESTION

Nobody predicts. They detect.

Current practice asks whether a valve *is* failing. It never asks whether it *will*.

WHAT A CARDIOLOGIST DOES TODAY

Opens the echo, compares three numbers against the patient's own baseline. **Gradient risen 10 or more, now 20 or more** means moderate deterioration. Otherwise: nothing wrong, return in a year. A threshold, not a prediction.

WHAT WE ASK

Can patient, prosthesis and serial echocardiography predict adjudicated deterioration **beyond valve age and type**, and beyond that threshold rule? Guideline surveillance stays. The model only says **who needs looking at sooner**.

Structural deterioration, with death competing

ELEMENT	DEFINITION
Time zero	The patient's own reference echo, 30 to 180 days after implant. Not a fixed day 90
Primary	First adjudicated VARC-3 stage 2 or 3 haemodynamic deterioration attributable to SVD
Key secondary	Bioprosthetic valve failure , the challenge's own phrase. Deterioration is primary because failure is a treatment decision: it is undercounted, and it arrives too late to change surveillance
Competing	Death; index-valve replacement for a non-SVD cause
Intercurrent	Treated thrombosis resolves, so the patient returns to risk
Cohort	3,400 to 5,100 patients by Riley's criteria, counted in patients and events, never in landmark rows

TWO RULES THAT DECIDE VALIDITY

A day-90 prediction cannot use an echo taken on day 150.

Anchoring on the actual reference study removes the contradiction and the leakage with it.

The echo that establishes an endpoint is never a predictor of that endpoint. Without this rule the model looks brilliant and is worthless.

The provided dataset, and what we got out of it

202

notes across 117 patients, the dataset we were given. All extraction results below come from it

97

implants documented: 53 surgical, 44 transcatheter

110

with valve model and generation named

1

patient has gradients in two different years

That last number is the finding. These records establish that extraction works. They cannot train a model: roughly twenty unadjudicated event mentions, year-only dates, almost no serial echo. The model is therefore developed on explicitly synthetic scenarios, and every figure says so.

Three extraction traps guarded in tests: **Valve in Valve: No** is a structured field, not an event. **Reoperation: No previous surgeries** likewise. Native pre-operative gradients sit in the same note as prosthetic ones, so a naive series is two valves interleaved.

A pipeline from a written note to a number a clinic can act on

1 · INGEST Operative and progress notes, echo reports, labs with LOINC, medications with RxNorm, procedure and death codes	2 · EXTRACT Rules over a 42-model device table from public FDA identifiers, then an LLM pass. Evidence spans kept	3 · ADJUDICATE VARC-3 staged against the patient's own reference echo. Stages 0 to 3 plus uncertain	4 · MODEL Landmark dataset, penalised cause-specific Cox for three causes, combined to cumulative incidence	5 · SERVE Two APIs and a clinic view. Three probabilities, drivers, reliability
--	---	--	---	---

PREPROCESSING, THE PART NOBODY DEMOS

- Canonical device, class, generation and size from free text such as **#21 Trifecta** or **Size: Ultra 26mm**
- Native versus prosthetic gradients separated by context; structured fields parsed by value, not phrase
- Echoes aligned to implant, reference study and landmark windows defined, stale values flagged at 18 months
- Medications reconstructed into dated exposure timelines by class

MODELS

- **Primary:** penalised cause-specific proportional hazards, splines, route-stratified baselines, for deterioration, death and non-SVD replacement
- **Challenger:** gradient-boosted survival, promoted only on calibration and Brier
- **Comparators:** null, valve age and type, the guideline threshold rule
- **Modules:** biomarker blocks and anticoagulant exposure, each measured for incremental value

One screen, three messages that are never merged

ASSESS NOW

A current abnormality on this echo. A **finding**, not a prediction

LOOK SOONER

Predicted 12-month risk high enough to bring the next assessment forward

OVERDUE

A scheduled echo that has not happened. A reminder needing no model

RELIABILITY

Device evidence, completeness, staleness. It can **withhold** a prediction

ON THE SCREEN BESIDE THEM

- The **three probabilities** at 12 months and 5 years: deterioration before death, death before deterioration, alive with the valve intact
- The patient's echo trajectory against the reference study, and the VARC-3 stage the module proposes
- The **main drivers**, labelled model sensitivities rather than treatment effects
- The guideline schedule, displayed and never modified
- The guideline rule's own verdict beside ours, and the Cox against boosting difference

Streamlit front end over two FastAPI services, extract and predict, running on Azure Container Apps.

Beating valve age proves nothing. Beat the rule.

One patient, 58, bicuspid valve, four years after a surgical implant.

MEASURE	DAY 90	YEAR 4
Mean gradient, mmHg	11	19
Dimensionless index	0.48	0.37
Effective orifice area, cm ²	1.7	1.25

Rose 8 mmHg. Stage 2 needs a rise of 10 to reach 20. He has crossed no line.

THE GAP

The rule says *nothing is wrong, return in a year.*

The model reads the trajectory and brings the assessment forward to six months.

Net benefit against that rule is the primary clinical comparison.
Illustrative values; the model shown is trained on synthetic scenarios.

Gains where information exists. Losses reported.

IPCW BRIER AT 5 YEARS	REF CORE + MODULES EVENTS			
Gradual stenotic	.0559	.0472	.0485	198
Biomarker, meaningful	.0524	.0476	.0468	178
Biomarker, absent	.0490	.0478	.0475	159
Irregular surveillance	.0639	.0602	.0584	189
Regurgitant, abrupt	.0776	.0782	.0790	267
High competing mortality	.0041	.0042	.0039	3

Lower is better. Reference is valve age and type alone, the weak bar. The guideline threshold rule has now been run on the same 10,701 landmarks: it returned an evaluable verdict on **0** of them, against **10,701** supported model predictions. It abstains without a complete severity assessment. Net benefit therefore remains unmeasured; see [docs/comparison/](#).

WHERE IT WORKS

Gradual stenotic: **AUC 0.80 (0.77–0.83)**, calibration slope **0.88**, IPA **0.21**. Removing serial echo takes Brier from .0472 to **.0575** — worse than the reference. **Updating is the whole gain.**

WHERE THE MODULES STAY QUIET

The biomarker block improves Brier when the scenario contains information and does nothing when it does not. That is the behaviour we specified in advance.

GRADIENT BOOSTING, THE PRE-SPECIFIED CHALLENGER

Same rows, same folds, 200 bootstraps by patient. Mean Brier difference **+5.7e-4**, interval **[-4.0e-6, +1.1e-3]**, spanning zero at every ladder step. Decision: **retain the Cox model**. The challenger failed its own promotion rule and we publish that.

WHERE IT LOSES, SAID FIRST

On abrupt regurgitant failure the reference beats us: no preceding

A live service, and an order we will not skip

LIVE PROTOTYPE

kairos-demo.blackground-92784e60
.swedencentral.azurecontainerapps.io

Note in, valve passport out with evidence spans. VARC-3 staging against the reference study. Three probabilities, drivers, reliability statement.

REPRODUCIBLE

make test && make quick

Extraction, six scenarios, training, evaluation. Privacy scan blocks any patient-level row from reaching version control.

1 · RETROSPECTIVE

Development in cohorts with adjudicated endpoints

2 · FREEZE

Temporal and external validation, untouched centre

3 · SILENT

Prospective, no clinical action on the output

4 · ONLY THEN

A trial of whether intervals can safely change

An observational model cannot show that looking *less* often is safe. So we never say it. It says *who to look at sooner*.